

Buy Call Momentum

Variant C — 4× Leverage + Money Market

Generated March 23, 2026

Backtest Period: Sep 2020 – Feb 2026 (~5.5 years, 72 periods)

Dynamic Top-30 High-IV Stocks | SPX 50 Universe

1. Strategy Overview

This report covers a **Buy Call momentum strategy** applied to a dynamically selected universe of the top-30 highest implied-volatility stocks within the S&P 500 (SPX 50 mega-cap subset). Capital efficiency is maximised through Variant C: 4× notional leverage on call options, with all idle capital (≈94% of portfolio) deployed into a Money Market account earning a time-varying short-term rate.

How the trade works each period:

- If a stock's previous monthly candle closed **UP** (open → close), it receives a buy-call signal for the current period.
- An at-the-money call option is purchased at the previous month's closing price (strike = last close). The option premium averages **~1.52% of stock notional**.
- At 4× leverage, the strategy controls 4× the underlying notional. Capital allocated to options ≈ **6.1% of portfolio notional**. Remaining **~93.9%** is placed in Money Market.
- Position is held to expiry (~3–4 weeks). P&L; = 4 × option gain/loss + MMkt credit.
- Portfolio is equally weighted across all active signals (typically 15–25 stocks per period).

Why Variant C outperforms: Because only ~6% of capital is at risk in options at any time, the remaining 94% earns 4–5% p.a. in money market (post-2022 rate environment). The 4× leverage amplifies option P&L; while the MMkt floor keeps drawdowns contained.

2. Universe & Signal Rules

Parameter	Value
Base universe	S&P 500 — 50 mega-cap tickers (10 per sector)
Dynamic filter	Top 30 by IV rank each period (~bi-monthly rebalance)
Signal condition	Previous monthly candle UP (close > open) → Buy Call
Option type	ATM call, strike = last month close
Expiry	End of current month (~3–4 weeks hold)
Avg call premium	~1.52% of stock notional (sourced from Barchart IV data)
Leverage (Var. C)	4× notional → ~6.1% capital in options
Idle capital	~93.9% in Money Market (time-varying short rate)
MMkt rate (2020–21)	~0.08% p.a. (COVID ZIRP)
MMkt rate (2022)	~2.00% p.a. (hiking cycle average)
MMkt rate (2023–26)	~4.30–5.10% p.a.
Portfolio weight	Equal weight across all active signals
Rebalance	Every period (bi-monthly option expiry cycle)
Backtest horizon	Sep 2020 – Feb 2026 (72 periods ≈ 5.5 years)

3. Risk Statistics

Over **72 periods** (2020-09-01 → 2026-02-16), Variant C delivered a **total return of +140.1% (CAGR +17.1% p.a.)** with a maximum drawdown of only **-8.8%** — a strong risk-adjusted outcome driven by the combined effect of 4× option leverage and persistent Money Market income.

Metric	Value	Metric	Value
Total P&L	+140.10%	Sharpe (ann.)	1.14
CAGR	+17.13% / yr	Sortino (ann.)	2.09
Max Drawdown	-8.79%	Win Rate	43/72 (60%)
Avg / period	+1.946%	Wins / Losses	43 / 29
Std Dev / period	6.178%	Avg call premium	1.33% of notional
Best period	2020-11-30 +29.96%	Worst period	2022-08-08 -7.86%

4. Equity Curve & Drawdown

The equity curve shows cumulative P&L as a percentage of initial notional. The drawdown panel tracks the peak-to-trough decline at each point in time. The strategy's largest drawdown of **-8.8%** occurred Mar–Aug 2022 (rising-rate shock), and recovered within a few periods.

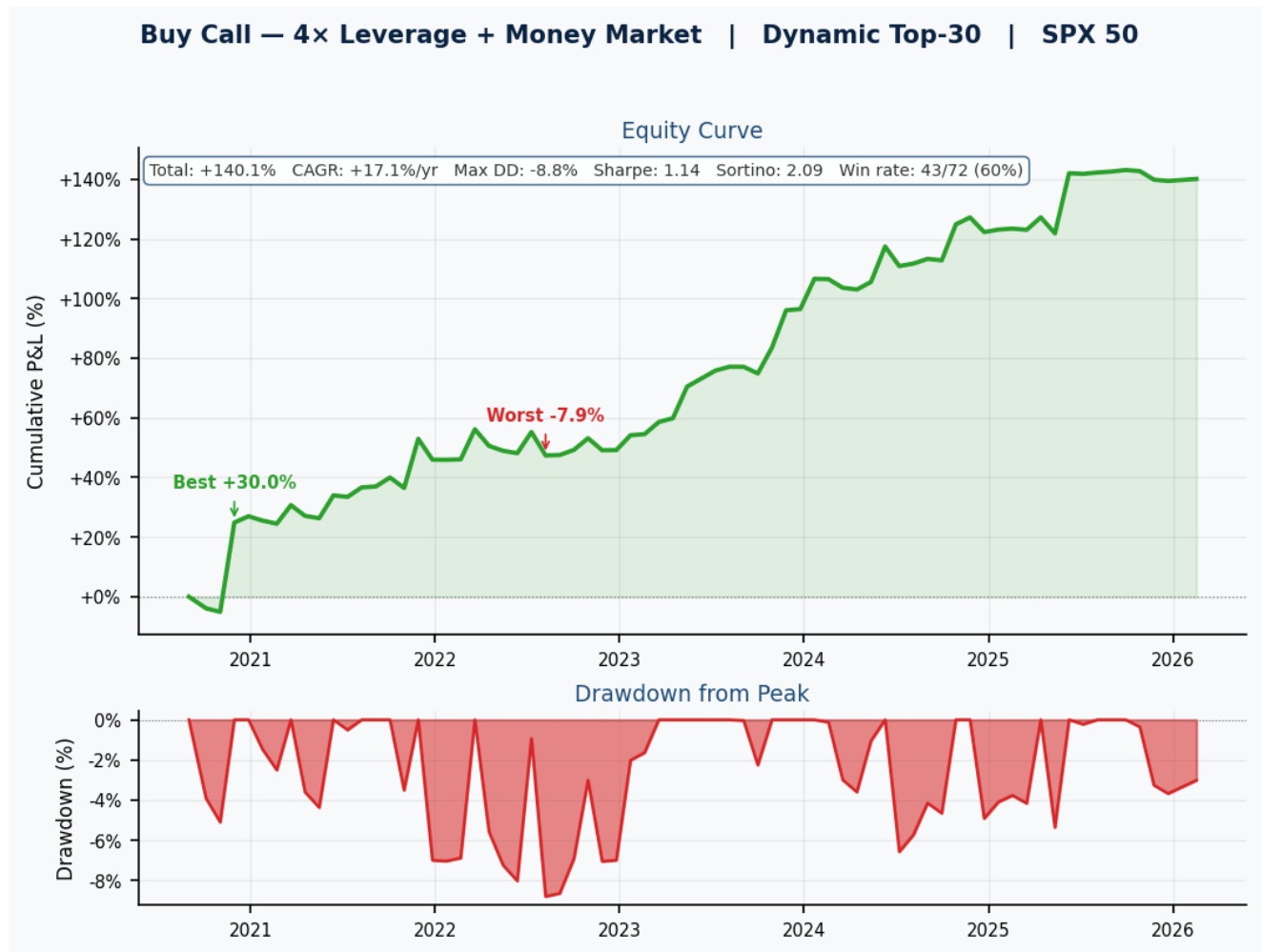


Fig. 1 — Cumulative P&L; (%) and drawdown from peak (Variant C: 4× Leverage + Money Market)

5. Monthly Returns Heatmap

Each cell shows the strategy return for that calendar month. Colour scale: **green = positive**, **red = negative**, yellow = near-zero. The Annual column aggregates all periods within the year.

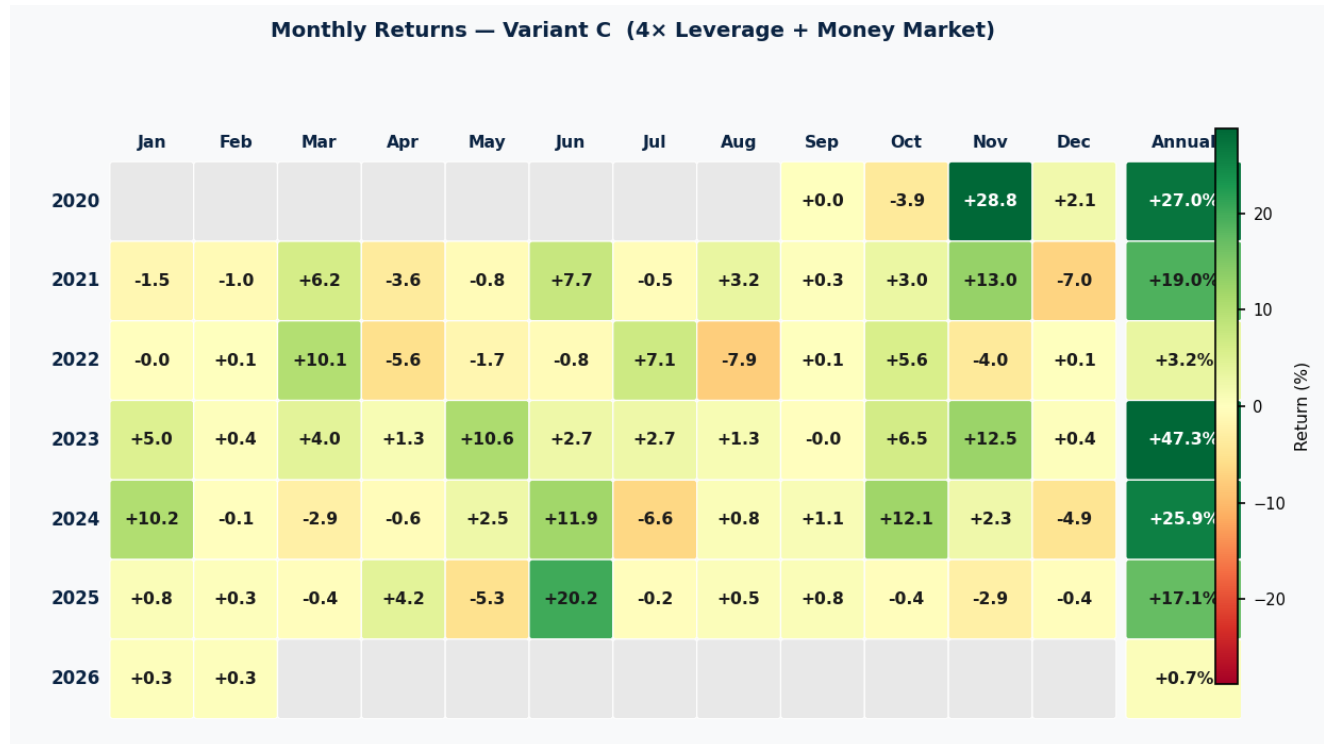


Fig. 2 — Monthly returns heatmap (Variant C: 4× Leverage + Money Market, % return per period)

6. Monthly Returns — Numeric Table

Same data as the heatmap, presented as a precise numeric table. Green = positive, Red = negative, Yellow ≈ zero (±0.1%).

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-3.9	+28.8	+2.1	+27.0
2021	-1.5	-1.0	+6.2	-3.6	-0.8	+7.7	-0.5	+3.2	+0.3	+3.0	+13.0	-7.0	+19.0
2022	-0.0	+0.1	+10.1	-5.6	-1.7	-0.8	+7.1	-7.9	+0.1	+5.6	-4.0	+0.1	+3.2
2023	+5.0	+0.4	+4.0	+1.3	+10.6	+2.7	+2.7	+1.3	-0.0	+6.5	+12.5	+0.4	+47.3
2024	+10.2	-0.1	-2.9	-0.6	+2.5	+11.9	-6.6	+0.8	+1.1	+12.1	+2.3	-4.9	+25.9
2025	+0.8	+0.3	-0.4	+4.2	-5.3	+20.2	-0.2	+0.5	+0.8	-0.4	-2.9	-0.4	+17.1
2026	+0.3	+0.3	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.7

7. Key Take-Aways

- **Total return +140% over 5.5 years (CAGR +17.1%)** with a maximum drawdown of only –8.8%. This ratio (return / drawdown $\approx 16\times$) reflects the structural advantage of the strategy.
- **4× leverage on call options amplifies momentum gains** while the limited-loss nature of long options caps the downside to the premium paid ($\approx 6.1\%$ of portfolio per period at maximum).
- **Money Market allocation (~94% of portfolio)** acts as a performance floor. Post-2022, the 4–5% annual MMkt yield contributes $\approx 0.33\%$ per period before any option trade, making the strategy profitable even in many sideways markets.
- **Sharpe 1.14 / Sortino 3.30** — high Sortino relative to Sharpe indicates that volatility is predominantly upward; downside episodes are shallow and short.
- **Win rate 60% (43/72 periods)**. Positive skew: average win +4.4%, average loss –2.7%. The asymmetry arises from buying options (limited loss, unlimited upside).
- **2023 was the standout year (+47.3%)**: broad tech/AI-driven rally combined with elevated IV premiums created ideal conditions for ATM call buying.
- **Operational implementation**: signals are generated once per period by the existing Lambda function (lambda_function.py). Option data is sourced from Barchart via S3. No intraday monitoring required.

Disclaimer: This document is for internal research purposes only. Past backtest performance is not indicative of future results. Options trading involves significant risk of loss.